

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing,

a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

ANSWER:

Scenario: Event Communication Security Using Caesar Cipher

A **Caesar Cipher** is a classical encryption technique in which each letter of the plaintext is shifted by a fixed number of positions in the alphabet.

Given:

- **Plaintext:** MEET AT NOON
- **Shift Key:** 4

Encryption Process

Each letter is shifted **4 positions forward**.

Plaintext M E E T A T N O O N

Ciphertext Q I I X E X R S S R

Encrypted Message (Ciphertext):

QIIX EX RSSR

Decryption Process

The authorized member knows the shift key is 4. To retrieve the original message, each letter is shifted **4 positions backward**.

Ciphertext Q I I X E X R S S R

Plaintext M E E T A T N O O N

Decrypted Message (Plaintext):

MEET AT NOON

Explanation

- During **encryption**, each letter in the message is shifted **4 positions forward** in the alphabet.

- During **decryption**, the receiver shifts each letter **4 positions backward** using the same key.
- Since the authorized member knows the shift value (**4**), they can successfully recover the original confidential message.

Final Answer

- **Plaintext:** MEET AT NOON
- **Encryption Key:** 4
- **Encrypted Message:** QIIX EX RSSR
- **Recovered Original Message:** MEET AT NOON

2. Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple

substitution mechanism rather than complex encryption.

Deduce the most likely original message.

Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

ANSWER:

Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message:

Ciphertext: GSRH RH Z HVXIVG

The message is most likely encrypted using the **Atbash Cipher**, a simple substitution cipher in which each letter of the alphabet is replaced by its opposite letter.

Atbash Mapping:

- A ↔ Z
- B ↔ Y
- C ↔ X
- D ↔ W
- E ↔ V
- F ↔ U
- G ↔ T
- H ↔ S
- I ↔ R
- J ↔ Q

- $K \leftrightarrow P$
- $L \leftrightarrow O$
- $M \leftrightarrow N$

Decryption

Ciphertext G S R H R H Z H V X I V G

Plaintext T H I S I S A S E C R E T

Original Message

THIS IS A SECRET

Justification

- The message follows the **Atbash Cipher**, where each letter is replaced with its opposite in the alphabet ($A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X$, etc.).
- The repeated pattern "RH" appears twice in the ciphertext and decrypts to "IS", which is a common English word.
- The single-letter word "Z" decrypts to "A", which is a valid English article.
- The last word "HVXIVG" decrypts to "SECRET", forming a meaningful sentence.

Final Answer

- **Ciphertext:** GSRH RH Z HVXIVG
- **Cipher Used:** Atbash Cipher (Simple Substitution Cipher)
- **Decrypted Message:** THIS IS A SECRET

3. Scenario: Secure Messaging System A company deploys a keyword-based encryption technique

for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure

confidentiality.

Demonstrate how the encryption process transforms the message step-by-step.

Explain how the repeating keyword influences each character of the ciphertext.

ANSWER:

Scenario: Secure Messaging System Using Vigenère Cipher

The company uses the Vigenère Cipher, a keyword-based encryption technique. In this method, each letter of the plaintext is encrypted using the corresponding letter of a repeating keyword.

Given

- Plaintext: HELLO
- Keyword: KEY

Step 1: Repeat the Keyword

Repeat the keyword until it matches the length of the plaintext.

Plaintext H E L L O

Keyword K E Y K E

Step 2: Convert Letters to Numbers

Using A = 0, B = 1, ..., Z = 25

Letter H E L L O

Value 7 4 11 11 14

Keyword Letter K E Y K E

Value 10 4 24 10 4

Step 3: Encrypt Each Letter

Formula:

Ciphertext = (Plaintext + Keyword) mod 26

Plaintext	Value	Keyword	Value	Calculation	Cipher Value	Cipher Letter
H	7	K	10	$(7 + 10) \bmod 26 = 17$	17	R
E	4	E	4	$(4 + 4) \bmod 26 = 8$	8	I
L	11	Y	24	$(11 + 24) \bmod 26 = 9$	9	J
L	11	K	10	$(11 + 10) \bmod 26 = 21$	21	V
O	14	E	4	$(14 + 4) \bmod 26 = 18$	18	S

Encrypted Message (Ciphertext)

RIJVS

Explanation

- The keyword "KEY" is repeated as KEYKE to match the length of the plaintext HELLO.

- Each letter of the keyword determines how many positions the corresponding plaintext letter is shifted.
 - Since the keyword repeats, the amount of shift changes for each character, making the encryption stronger than a simple Caesar cipher.
 - Only someone who knows the same keyword (KEY) can correctly decrypt the ciphertext back to HELLO.
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Final Answer

- Plaintext: HELLO
- Keyword: KEY
- Repeated Keyword: KEYKE
- Encryption Technique: Vigenère Cipher
- Encrypted Message (Ciphertext): RIJVS

4. Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file

appears normal upon casual inspection.

Analyze how investigators could detect the presence of hidden data within the image.

Recommend two techniques that could strengthen the security of such hidden communications against

detection or extraction.

ANSWER:

Scenario: Suspicious Digital Asset Investigation

Investigators suspect that a company logo image contains hidden confidential information. This is an example of image steganography, where secret data is embedded inside an image without visibly changing its appearance.

Detection of Hidden Data

Investigators can detect the presence of hidden data using the following methods:

1. **Steganalysis**
 - Specialized steganalysis tools analyze the image for unusual patterns or modifications.
 - They examine pixel values, metadata, and file structure to identify signs of hidden information.

2. Statistical and Pixel Analysis

- Investigators compare the image's pixel values and color distribution with those of a normal image.
 - Abnormal changes, especially in the Least Significant Bits (LSBs) of pixels, may indicate embedded data.
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Techniques to Strengthen Security

1. Encrypt the Secret Data Before Embedding

- Encrypt the confidential message using a strong encryption algorithm (such as AES) before hiding it in the image.
- Even if the hidden data is extracted, it cannot be understood without the correct encryption key.

2. Use Advanced Steganography Techniques

- Instead of embedding data in predictable pixel locations, use randomized or adaptive embedding methods controlled by a secret key.
 - This makes the hidden data much harder to detect or extract through steganalysis.
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Conclusion

By using steganalysis and statistical pixel analysis, investigators can determine whether an image contains hidden information. To improve security, organizations should encrypt the data before embedding and use advanced, key-based steganography techniques, making hidden communications more resistant to detection and extraction.

Final Answer

- **Detection Methods:**
 1. Steganalysis using specialized tools.
 2. Statistical and pixel (LSB) analysis.
- **Security Enhancement Techniques:**
 1. Encrypt the secret data before embedding.
 2. Use advanced key-based/randomized steganography techniques.

5. Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a

transposition-based encryption method with three levels of arrangement. A received message appears as:

“WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

ANSWER:

Scenario: Legacy Encryption in Corporate Communication

The given ciphertext is:

Ciphertext: WECRLTEERDSOEFEAOCAIVDEN

The message is encrypted using a Rail Fence Cipher (3 Rails), which is a transposition-based encryption technique.

a. Determine the Original Message

Using Rail Fence Cipher Decryption (3 Rails), the ciphertext is rearranged to recover the original plaintext.

Original Message (Plaintext):

WE ARE DISCOVERED FLEE AT ONCE

b. Why is this Method Vulnerable in Modern Communication Systems?

The Rail Fence Cipher is considered insecure because:

1. **No Secret Key**
 - The encryption only rearranges the positions of letters without changing them.
 - If an attacker knows the algorithm and the number of rails, the message can be easily decrypted.
2. **Easy to Break**
 - Since there are only a few possible rail arrangements, attackers can use brute-force methods to recover the plaintext quickly.
3. **No Protection Against Modern Attacks**
 - Frequency analysis, pattern recognition, and automated cryptanalysis tools can easily detect and decrypt the message.

4. Provides Weak Confidentiality

- It does not use complex mathematical operations or strong encryption keys, making it unsuitable for protecting sensitive information.

c. Practical Enhancement

A practical way to improve confidentiality is to use AES (Advanced Encryption Standard) instead of the Rail Fence Cipher.

Advantages of AES:

- Uses strong secret keys (128-bit, 192-bit, or 256-bit).
- Highly resistant to brute-force and cryptanalysis attacks.
- Widely used for secure communication in modern systems.
- Provides strong confidentiality for sensitive corporate data.

Final Answer

a. Original Message

WE ARE DISCOVERED FLEE AT ONCE

b. Vulnerabilities

- No secret encryption key.
- Easy to decrypt using brute force.
- Vulnerable to pattern analysis and cryptanalysis.
- Does not provide strong security for modern communication.

c. Practical Enhancement

Replace the Rail Fence Cipher with AES (Advanced Encryption Standard) to ensure strong encryption and secure communication.